



# Differentiation · Mark Scheme

Chapter 7: Differentiation. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

## Section A – First principles

- 1 (a)  $\frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h}$  M1 =  $6 + h$  A1. (b) As  $h \rightarrow 0$ , chord gradient  $\rightarrow 6$  B1.
- 2  $\lim_{h \rightarrow 0} \frac{4(x+h)^2 - 4x^2}{h}$  M1;  $= \lim_{h \rightarrow 0} \frac{8xh + 4h^2}{h} = \lim_{h \rightarrow 0} (8x + 4h)$  M1;  $= 8x$  A1.
- 3 Correct set-up M1; numerator  $2xh + h^2 - 5h$  A1;  $\lim(2x + h - 5)$  M1;  $= 2x - 5$  A1.
- 4 Set-up M1;  $(x+h)^3 - x^3 = 3x^2h + 3xh^2 + h^3$  A1; divide by  $h$  M1; limit  $3x^2$  A1.
- 5  $\frac{\sin(x+h) - \sin x}{h}$  M1;  $= \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$  M1;  $= \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}$  M1 A1; as  $h \rightarrow 0$ :  $\sin x(0) + \cos x(1) = \cos x$  A1.

## Section B – Polynomials, tangents and normals

- 6  $12x^2 - 10x + 7$  M1 A1.
- 7  $y = 6x^{\frac{1}{2}} + 8x^{-2} - \frac{1}{3}x^{-1}$  M1;  $3x^{-\frac{1}{2}} - 16x^{-3} + \frac{1}{3}x^{-2}$  A1 A1.
- 8  $y = x^{\frac{3}{2}} + 3x^{\frac{1}{2}}$  M1;  $y' = \frac{3}{2}x^{\frac{1}{2}} + \frac{3}{2}x^{-\frac{1}{2}}$  A1; at  $x = 4$ : 3.75 A1.
- 9  $y(2) = 5$ ,  $y' = 3x^2 - 2 = 10$  M1 A1; normal gradient  $-\frac{1}{10}$  M1;  $x + 10y - 52 = 0$  A1.
- 10(a)  $3x^2 - 12x + 9$  M1 A1. 10(b)  $P(4, 5)$ , gradient 9 M1 A1;  $y = 9x - 31$  A1.
- 10(c)  $3x^2 - 12x + 9 = 9 \Rightarrow 3x(x - 4) = 0$  M1 A1;  $Q(0, 1)$  A1. 10(d) Gradient  $-\frac{1}{9}$  M1;  $y = -\frac{1}{9}x + 1$  A1.

## Section C – Stationary points

- 11  $2x - 6 = 0$  M1;  $(3, -5)$  A1; minimum (positive  $x^2$  coefficient or  $y'' = 2 > 0$ ) B1.
- 12  $3x^2 - 6x - 9 = 0 \Rightarrow x = 3, -1$  M1 A1;  $(3, -22)$ ,  $(-1, 10)$  A1;  $y'' = 6x - 6$ :  $y''(3) = 12 > 0$  min,  $y''(-1) = -12 < 0$  max M1 A1.
- 13  $3x^2 - 12 > 0$  M1; critical values  $\pm 2$  A1;  $x < -2$  or  $x > 2$  A1.
- 14  $f'(x) = 3x^2 + 6x + 5$  B1;  $= 3(x+1)^2 + 2$  M1;  $> 0$  for all  $x$ , so increasing A1.
- 15(a)  $y' = 2 - 16x^{-3}$  M1 A1;  $x^3 = 8 \Rightarrow x = 2$  A1;  $(2, 6)$  A1.
- 15(b)  $y'' = 48x^{-4}$  M1;  $y''(2) = 3 > 0$ , minimum A1. 15(c)  $k > 6$  M1 A1.

## Section D – Inflection

- 16  $f' = 4x^3 - 6x^2$  M1;  $f'' = 12x^2 - 12x$  A1.
- 17  $y'' = 6x - 12 = 0 \Rightarrow x = 2$  M1;  $y''$  changes sign (e.g.  $-0.6, +0.6$  at 1.9, 2.1) M1;  $(2, -11)$  A1.
- 18  $y'' = 6x - 6$  M1; concave for  $x < 1$  A1; convex for  $x > 1$  A1.
- 19  $y'' = 12x^2 - 24x = 12x(x - 2)$  M1;  $x = 0, 2$  A1; sign change at both M1;  $(0, 0)$  and  $(2, -16)$  A1;  $y'(0) = 0$ , so  $(0, 0)$  is the stationary point of inflection ( $y'(2) = -16 \neq 0$ ) A1.
- 20(a)  $f' = 15x^4 - 20x^3$ ,  $f'' = 60x^3 - 60x^2$  M1 A1;  $= 60x^2(x - 1)$ , zero at  $x = 0, 1$  A1.
- 20(b)  $f''(0.9) < 0$ ,  $f''(1.1) > 0$  M1; sign change, so inflection A1;  $(1, -2)$  A1.



**20(c)**  $f''$  is negative on both sides of  $x = 0$  (because of the  $x^2$  factor) *M1*; no change of concavity, so not an inflection *A1*.

## Section E – Exponentials and logarithms

**21**  $3e^x + \frac{4}{x}$  *B1 B1*.

**22**  $5e^{5x} - \frac{1}{x}$  *B1 B1*. Note:  $\ln(2x) = \ln 2 + \ln x$ , so its derivative is  $\frac{1}{x}$ .

**23**  $\frac{dy}{dx} = 2^x \ln 2$  *M1 A1*; at  $x = 3$ :  $8 \ln 2$  *A1*.

**24**  $2e^{2x} - 6 = 0 \Rightarrow x = \frac{1}{2} \ln 3$  *M1 A1*;  $y = 3 - 3 \ln 3$  *A1*;  $y'' = 4e^{2x} > 0$ , minimum *B1*.

**25(a)**  $\frac{2}{x} - 1 = 0$  *M1*;  $x = 2$  *A1*;  $(2, 2 \ln 2 + 3)$  *A1*.

**25(b)**  $y'' = -\frac{2}{x^2} < 0$  *M1*; maximum *A1*. **25(c)**  $y(1) = 4$ , gradient 1 *M1 A1*;  $y = x + 3$  *A1*.

## Section F – Trigonometric functions

**26**  $3 \cos x + 2 \sin x$  *B1 B1*.

**27**  $4 \cos 4x - \frac{1}{2} \sin\left(\frac{x}{2}\right)$  *B1 B1*.

**28**  $5 \sec^2 x - 2$  *M1 A1*; at  $\frac{\pi}{4}$ :  $5(2) - 2 = 8$  *A1*.

**29**  $1 - 2 \sin x = 0 \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$  *M1 A1*;  $\left(\frac{\pi}{6}, \frac{\pi}{6} + \sqrt{3}\right), \left(\frac{5\pi}{6}, \frac{5\pi}{6} - \sqrt{3}\right)$  *A1*;  $y'' = -2 \cos x$ : max at  $\frac{\pi}{6}$ , min at  $\frac{5\pi}{6}$  *M1 A1*.

**30(a)**  $2 \cos x + 2 \cos 2x$  *B1 B1*.

**30(b)**  $2 \cos x + 2(2 \cos^2 x - 1) = 0$  *M1*;  $(2 \cos x - 1)(\cos x + 1) = 0$  *M1*;  $x = \frac{\pi}{3}, \frac{5\pi}{3}$  *A1*;  $x = \pi$  *A1*.

**30(c)** Gradient at  $x = 0$  is 4 *M1*;  $y = 4x$  *A1*.

## Section G – Chain rule

**31**  $15(3x+1)^4$  *M1 A1*.

**32**  $\frac{1}{2}(x^2+5)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{x^2+5}}$  *M1 A1*; at  $x = 2$ :  $\frac{2}{3}$  *A1*.

**33** (a)  $2xe^{x^2}$  *M1 A1*; (b)  $\frac{2x}{x^2+1}$  *B1*.

**34** (a)  $3 \sin^2 x \cos x$  *M1 A1*; (b)  $-3x^2 \sin(x^3)$  *B1*.

**35(a)**  $\frac{1}{2}(4x+1)^{-\frac{1}{2}} \cdot 4 = \frac{2}{\sqrt{4x+1}}$  *M1 A1*.

**35(b)** At  $x = 2$ :  $y = 3$ , gradient  $\frac{2}{3}$  *B1*; normal gradient  $-\frac{3}{2}$  *M1*;  $3x + 2y - 12 = 0$  *A1*.

**35(c)**  $A(4, 0), B(0, 6)$  *M1 A1*; area = 12 *A1*.

## Section H – Product rule

**36**  $2xe^x + x^2e^x = xe^x(x+2)$  *M1 A1*.

**37**  $3x^2 \sin x + x^3 \cos x$  *M1 A1*.

**38**  $\ln x + \frac{x+1}{x}$  *M1 A1*; at  $x = 1$ :  $0 + 2 = 2$  *A1*.

**39**  $(2x-1)^4 + x \cdot 4(2x-1)^3 \cdot 2$  *M1 A1*;  $= (2x-1)^3[(2x-1) + 8x]$  *M1*;  $= (2x-1)^3(10x-1)$  *A1*.

**40(a)**  $2xe^{-x} - x^2e^{-x}$  *M1 A1*;  $= e^{-x}(2x-x^2)$  *A1*.



40(b)  $e^{-x} \neq 0$ , so  $x(2-x) = 0$  M1; (0,0) A1; (2,  $4e^{-2}$ ) A1.

40(c)  $f''(0) = 2 > 0$ : minimum B1;  $f''(2) = -2e^{-2} < 0$ : maximum B1.

## Section I – Quotient rule

41  $\frac{(x+1)-x}{(x+1)^2} = \frac{1}{(x+1)^2}$  M1 A1.

42  $\frac{x \cdot 2e^{2x} - e^{2x}}{x^2}$  M1 A1;  $= \frac{e^{2x}(2x-1)}{x^2}$  A1.

43  $\frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x}$  M1;  $= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$  A1;  $= \frac{1}{\cos^2 x} = \sec^2 x$  A1.

44  $\frac{2x(x-1) - (x^2+3)}{(x-1)^2}$  M1 A1;  $= \frac{x^2-2x-3}{(x-1)^2} = 0$  M1;  $x = 3$  or  $-1$  A1; (3,6) and  $(-1,-2)$  A1.

45(a)  $\frac{x \cdot \frac{1}{x} - \ln x}{x^2}$  M1 A1;  $= \frac{1 - \ln x}{x^2}$  A1.

45(b)  $\ln x = 1 \Rightarrow x = e$  M1;  $(e, \frac{1}{e})$  A1.

45(c) Quotient (or product) rule on  $f'(x)$  M1;  $\frac{-x-2x(1-\ln x)}{x^4} = \frac{2\ln x-3}{x^3}$  A1;  $f''(e) = -\frac{1}{e^3} < 0$ , maximum A1.

## Section J – Implicit differentiation

46  $2x + 2y \frac{dy}{dx} = 0$  M1;  $\frac{dy}{dx} = -\frac{x}{y}$  A1.

47  $2x + 3y + 3x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$  M1 A1;  $\frac{dy}{dx} = -\frac{2x+3y}{3x+2y}$  A1.

48  $3x^2 + 3y^2 \frac{dy}{dx} = 0$  M1 A1; at (2,1):  $-4$  A1.

49  $9 - 6 + 4 = 7$  B1;  $2x - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$  M1; gradient  $\frac{y-2x}{2y-x} = -4$  A1;  $y = -4x + 14$  A1.

50(a)  $2x + 2y + 2x \frac{dy}{dx} - 6y \frac{dy}{dx} = 0$  M1 M1 A1;  $\frac{dy}{dx} = \frac{x+y}{3y-x}$  A1.

50(b)  $x + y = 0 \Rightarrow y = -x$  M1;  $x^2 - 2x^2 - 3x^2 + 16 = 0$  M1;  $x = \pm 2$  A1; (2, -2) and  $(-2, 2)$  A1.

## Section K – Parametric differentiation

51  $\frac{4}{2t} = \frac{2}{t}$  M1 A1.

52  $\frac{2 \cos \theta}{-3 \sin \theta} = -\frac{2}{3} \cot \theta$  M1 A1; at  $\frac{\pi}{4}$ :  $-\frac{2}{3}$  A1.

53  $\frac{dx}{dt} = 1 - t^{-2}$ ,  $\frac{dy}{dt} = 1 + t^{-2}$  M1;  $\frac{dy}{dx} = \frac{1+t^{-2}}{1-t^{-2}}$  A1; at  $t = 2$ :  $\frac{5}{3}$  A1.

54 Point (18, 12) B1;  $\frac{dy}{dx} = \frac{4}{4t} = \frac{1}{t}$  M1 A1;  $x - 3y + 18 = 0$  A1.

55(a)  $\frac{dy}{dx} = \frac{3t^2-3}{2t}$  M1 A1 A1.

55(b)  $t = \pm 1$  M1; (0, -2) and (0, 2) A1 A1.

55(c) Point (3, 2), gradient  $\frac{9}{4}$  B1; normal gradient  $-\frac{4}{9}$  M1;  $4x + 9y - 30 = 0$  A1.



## Section L – Connected rates of change

- 56  $\frac{dy}{dt} = 3x^2 \cdot \frac{dx}{dt}$  M1; = 54 A1.
- 57  $\frac{dA}{dr} = 2\pi r = 10\pi$  B1;  $\frac{dr}{dt} = \frac{10}{10\pi}$  M1; =  $\frac{1}{\pi} \approx 0.318 \text{ cm s}^{-1}$  A1.
- 58  $\frac{dV}{dx} = 3x^2 = 48$  B1;  $\frac{dV}{dt} = 48 \times 0.5$  M1; =  $24 \text{ cm}^3 \text{ s}^{-1}$  A1.
- 59  $\frac{dV}{dr} = 4\pi r^2 = 100\pi$  M1 A1;  $\frac{dr}{dt} = \frac{50}{100\pi}$  M1; =  $0.159 \text{ cm s}^{-1}$  A1.
- 60(a)  $V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h$  M1; =  $\frac{\pi h^3}{12}$  A1.
- 60(b)  $\frac{dV}{dh} = \frac{\pi h^2}{4} = 4\pi$  M1 A1;  $\frac{dh}{dt} = \frac{8}{4\pi}$  M1; =  $\frac{2}{\pi} \approx 0.637 \text{ cm s}^{-1}$  A1.
- 60(c)  $\frac{dr}{dt} = \frac{1}{2} \frac{dh}{dt}$  M1; =  $\frac{1}{\pi} \approx 0.318 \text{ cm s}^{-1}$  A1.

## Section M – Optimisation

- 61  $P = x(20 - x)$  M1;  $P' = 20 - 2x = 0 \Rightarrow x = 10$  M1; max product 100 A1.
- 62 (a) Base  $(12 - 2x)$  square, height  $x$  B1. (b)  $V' = (12 - 2x)^2 - 4x(12 - 2x) = (12 - 2x)(12 - 6x)$  M1 A1;  $x = 2$  (reject  $x = 6$ ) A1;  $V = 128 \text{ cm}^3$  A1.
- 63 (a)  $h = \frac{250}{r^2}$  M1;  $S = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{500\pi}{r}$  A1. (b)  $S' = 4\pi r - \frac{500\pi}{r^2} = 0 \Rightarrow r = 5$  M1 A1;  $S = 150\pi \text{ cm}^2$  A1.
- 64 Sides  $x, x$  and  $60 - 2x$ :  $A = x(60 - 2x)$  M1 A1;  $A' = 60 - 4x = 0 \Rightarrow x = 15$  M1;  $A = 450 \text{ m}^2$  A1.
- 65(a)  $S = 2(2x^2) + 2(xh) + 2(2xh) = 4x^2 + 6xh = 600$  M1 A1;  $h = \frac{600 - 4x^2}{6x}$ ,  $V = 2x^2h = 200x - \frac{4}{3}x^3$  A1.
- 65(b)  $V' = 200 - 4x^2 = 0$  M1;  $x = \sqrt{50} = 5\sqrt{2}$  A1;  $V = \frac{2000\sqrt{2}}{3}$  M1; =  $943 \text{ cm}^3$  A1.
- 65(c)  $V'' = -8x$  M1;  $< 0$  at  $x = 5\sqrt{2}$ , so maximum A1.

## Roar-level challenge

- 66  $\ln y = x \ln x$  M1;  $\frac{1}{y} \frac{dy}{dx} = \ln x + 1$  A1;  $\ln x = -1 \Rightarrow x = e^{-1}$  M1; stationary point  $(e^{-1}, e^{-1/e})$  A1.